

Library Management System Database Design

1. Entity List with Attributes

The following table details the relational schema, data types, constraints, and operational descriptions for each entity within the library management system.

Entity	Attribute	Data Type	Constraints	Description
Publisher	publisher_id	INT	PRIMARY KEY	Unique publisher identifier
	name	VARCHAR (100)	NOT NULL	Publisher name
	address	VARCHAR (200)		Publisher address
	phone	VARCHAR (20)		Contact phone
Author	author_id	INT	PRIMARY KEY	Unique author identifier
	first_name	VARCHAR (50)	NOT NULL	Author's first name
	last_name	VARCHAR (50)	NOT NULL	Author's last name
	biography	TEXT		Author biography

Entity	Attribute	Data Type	Constraints	Description
Book	book_id	INT	PRIMARY KEY	Unique book identifier (title/edition)
	isbn	VARCHAR (13)	UNIQUE, NOT NULL	International Standard Book Number
	title	VARCHAR (200)	NOT NULL	Book title
	edition	INT		Edition number
	publication_year	YEAR		Year of publication
	publisher_id	INT	FOREIGN KEY → Publisher(publisher_id)	Publisher of this book
	description	TEXT		Brief description
Book_Author	book_id	INT	PRIMARY KEY, FOREIGN KEY → Book(book_id)	Composite PK
	author_id	INT	PRIMARY KEY, FOREIGN KEY → Author(author_id)	Composite PK

Entity	Attribute	Data Type	Constraints	Description
Category	category_id	INT	PRIMARY KEY	Unique category identifier
	name	VARCHAR (50)	NOT NULL, UNIQUE	Category name (e.g., Fiction, Science)
Book_Category	book_id	INT	PRIMARY KEY, FOREIGN KEY → Book(book_id)	Composite PK
	category_id	INT	PRIMARY KEY, FOREIGN KEY → Category(category_id)	Composite PK
Library_Branch	branch_id	INT	PRIMARY KEY	Unique branch identifier
	name	VARCHAR (100)	NOT NULL	Branch name (e.g., Central Library)
	location	VARCHAR (200)	NOT NULL	Address or area
	phone	VARCHAR (20)		Branch phone
Book_Copy	copy_id	INT	PRIMARY KEY	Unique copy identifier (per physical book)

Entity	Attribute	Data Type	Constraints	Description
	book_id	INT	FOREIGN KEY → Book(book_id), NOT NULL	Which title this copy belongs to
	branch_id	INT	FOREIGN KEY → Library_Branch(branch_id), NOT NULL	Owning branch
	status	ENUM('available', 'loaned', 'reserved', 'lost')	NOT NULL, DEFAULT 'available'	Current status
	acquisition_date	DATE		Date added to collection
Member	member_id	INT	PRIMARY KEY	Unique member identifier
	first_name	VARCHAR (50)	NOT NULL	Member first name
	last_name	VARCHAR (50)	NOT NULL	Member last name
	address	VARCHAR (200)		Postal address
	phone	VARCHAR		Contact phone

Entity	Attribute	Data Type	Constraints	Description
		(20)		
	email	VARCHAR (100)	UNIQUE	Email address
	membership_date	DATE	NOT NULL, DEFAULT CURRENT_DATE	Date joined
	membership_type	ENUM('regular', 'premium', 'staff')	NOT NULL, DEFAULT 'regular'	Type of membership
Loan	loan_id	INT	PRIMARY KEY	Unique loan transaction ID
	copy_id	INT	FOREIGN KEY → Book_Copy(copy_id), NOT NULL	Which copy is loaned
	member_id	INT	FOREIGN KEY → Member(member_id), NOT NULL	Who borrowed it
	loan_date	DATE	NOT NULL, DEFAULT CURRENT_DATE	Date borrowed
	due_date	DATE	NOT NULL	Calculated as loan_date +

Entity	Attribute	Data Type	Constraints	Description
				loan_period
	return_date	DATE		NULL if not yet returned
Fine	fine_id	INT	PRIMARY KEY	Unique fine ID
	loan_id	INT	FOREIGN KEY → Loan(loan_id), NOT NULL, UNIQUE	Each loan can have at most one fine
	amount	DECIMAL (6,2)	NOT NULL	Fine amount (e.g., 0.50 per day overdue)
	paid_status	BOOLEAN	NOT NULL, DEFAULT FALSE	Whether fine has been paid
	issue_date	DATE	NOT NULL, DEFAULT CURRENT_DATE	Date fine was issued
Reservation	reservation_id	INT	PRIMARY KEY	Unique reservation ID
	book_id	INT	FOREIGN KEY → Book(book_id), NOT NULL	Title being reserved

Entity	Attribute	Data Type	Constraints	Description
	member_id	INT	FOREIGN KEY → Member(member_id), NOT NULL	Member who reserved
	branch_id	INT	FOREIGN KEY → Library_Branch(branch_id), NOT NULL	Pickup branch
	reservation_date	DATETIME	NOT NULL, DEFAULT CURRENT_TIMESTAMP	When reservation was placed
	status	ENUM('pending', 'fulfilled', 'cancelled', 'expired')	NOT NULL, DEFAULT 'pending'	Current reservation state
	expiration_date	DATE	NOT NULL	Reservation expires after N days
Staff	staff_id	INT	PRIMARY KEY	Unique staff identifier
	first_name	VARCHAR(50)	NOT NULL	Staff first name

Entity	Attribute	Data Type	Constraints	Description
	last_name	VARCHAR (50)	NOT NULL	Staff last name
	role	VARCHAR (50)	NOT NULL	e.g., Librarian, Assistant
	branch_id	INT	FOREIGN KEY → Library_Branch(branch_id), NOT NULL	Branch where staff works

2. Relationships and Cardinalities

The structured interactions between the schema's entities are formalized below.

Relationship	From Entity	To Entity	Cardinality	Description
Publishes	Publisher	Book	1 : M	One publisher can publish many books. A book has exactly one publisher.
Writes	Author	Book	M : N	Many authors can write many books (via Book_Author).
Belongs to Category	Book	Category	M : N	A book can belong to

Relationship	From Entity	To Entity	Cardinality	Description
				several categories (via Book_Category).
Has Copies	Book	Book_Copy	1 : M	One book title can have many physical copies.
Located at Branch	Book_Copy	Library_Branch	M : 1	Many copies belong to a single branch.
Employs	Library_Branch	Staff	1 : M	One branch employs many staff members.
Borrows	Member	Loan	1 : M	One member can have many loans.
Loans Copy	Book_Copy	Loan	1 : M	A copy can be loaned many times (historically).
Incurs Fine	Loan	Fine	1 : 0..1	A loan may generate zero

Relationship	From Entity	To Entity	Cardinality	Description
				or one fine.
Places Reservation	Member	Reservation	1 : M	One member can place many reservations.
Reserves Book	Book	Reservation	1 : M	One book title can have many reservations.
Pickup Branch	Library_Branch	Reservation	1 : M	A branch can be pickup location for many reservations.

3. Additional Business Rules

The implementation of logic filters and validation standards governing the system data state includes:

- **Loan Constraints:** A loan's due_date must be greater than or equal to the loan_date.
- **Fine Logic:** A fine can only be created if the return_date evaluates to greater than the due_date (representing an overdue status).
- **Reservation Lifecycles:** A reservation is automatically cancelled if it is not fulfilled before its designated expiration_date.
- **Transactional Exclusivity:** A Book_Copy carrying a status equal to 'loaned' is locked from additional loan staging until its return state is captured.
- **Application Rule:** Any member carrying total unpaid fines greater than \$10 will be restricted from establishing new loan interactions.